

Exploratory Data Analysis | Singapore Total Fertility Rate & Total Live Births

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1 Introduction

This exploratory data analysis (EDA) investigates the temporal features of two significant population indicators for Singapore:

- **Total Fertility Rate (TFR):** the average number of children a woman would have over her lifetime at current age-specific fertility rates;
- **Total Live Births (TLB):** the total number of live births recognised per year.

The dates for the SingaporeCensus.csv reference dataset span from 1960 to 2024, sourced from the Singapore Department of Statistics. The objective is to recognise and analyse long-run trends, assess stationarity, and identify candidate time series models for a proposed training period 1960 - 2012 and a proposed forecasting performance assessed over the remaining 2013 - 2024 period.

2 Data Loading & Reshaping

The given data is in a CSV file, which is optimal for computation. However, the data is in a ‘wide format’, where its temporal variables (i.e. years) are given as columns instead of years. We transpose the data and extract the two data columns we are interested in.